System for Reaching Prime Numbers Trough Lines

Aaron Cuevas (Aaron-Cuevas)

2026-06-28

An insight looking trough prime numbers graphical arrangement

1 I

s possible to explain visually why some geometrical patterns appear in certain prime numer representations. Looking trough geometry it can be possible to explain the distribution of complex patterns. The course of ideas will follow until drawing over the known polar-spiral structure, lines, being the result of projecting residual classes.

1.1 T

he prime numbers are commonly represented with the equations of the polar-spiral graph, where every prime number with radio *p* and angle *p*:

$$z_p = pe^{ip} \tag{1}$$

And this behaviour come from a polar representation of integers, with a gap made between certain integers and the distribution of prime numbers in residual clases coprimes of module (N)

1.2 E

very natural number can be colocated in the plane trough [z$_n = ne^{in}]Thisequalsassaying[x_n =$
$n\cos(n)][y_n = n\sin(n)]Wherethenumber(n)determinessimultaneouslythedistancetocenterandtherotationangle.$
$[z_p = pe^{ip}]Where(p)passesbytheprimenumbers.$

2 Partitions on the spiral

For concreting the multiple-partitions spiral structure, we choose a module (N), reaching that every natural number can be calssified according to its residue. [n $\equiv r \pmod{N})]Andit'sfoundthatnumbersofoneresidu$
$r+kN]Puttingtheninanpolar-spiral[z_{r+kN} = (r+kN)e^{i(r+kN)}]Andseparatingtheangularterms:$
$[z_{r+kN} = e^{ir}(r+kN)e^{ikN}]Stabishingtherelationswiththenearbymultiples, wehavethatif(N)isnearamultipleof(2$
$[N \approx 2\pi m]andbythat: [e^{ikN} \approx 1]Impliesthatthenumbersofonesameresidualclasstendstobenearthesameangular$

2.1 Introduction of Dirichet Theorem

The tendency of the residual classes for obtaining infinite prime numbers don't apply to all of them.
If a residue (r) shares a factor with (N), so a lot of numbers of the form [r+kN] Will be compound.
For that are considered specially the classes where [$\gcd(r, N) = 1]Beingthemtheresidualcoprimalclasseswith(N).$

$[r, r+N, r+2N, r+3N, \ldots]Thelinescorrespondstoresidualclasses(r \pmod N)), with[\gcd(r, N) = 1]$

2.2 Module 44

Remembering the example given by the initial explanation where is inspired this method, we have that the number 44 is near of seven whole spins, $[\,44 \approx 7(2\pi)]Forthattheresidualclassesofmodule(44)generatesmo$ $1]as : [\varphi(44) = 20]Therearetwentypermitedclassesthatcorrespondsto20linesexpectedinthevisualization.$

2.3 Error Tracking

As the alineation ain't perfect because (N) doesn't equal exactly a multiple of $(2\pi).Itcanbewrited[N = 2\pi m+\varepsilon]Where(\varepsilon)istheangularerror.Everytimewestepaheadoneunityinthesameresidualclass, theerroracumulat$ $[k\varepsilon]If(\varepsilon)islittle, thelineseemsstraightduringverylongtime.Ifitslarger, thelinebendsfaster.Soon, thestraightnes$ $[N \approx 2\pi m]$

/sectionProposed method Formulating the analysis method, we find that, in first, we graph whe prime numbers through the function: $[\,\mathrm{z}_p = pe^{ip}]Then, wechooseamodulo(N)nearamultipleof(2\pi) :$ $[N \approx 2\pi m]Andwedividethenumbersinresidualclasses : [p \equiv r \pmod N)]Weidentifythecoprimalclasses :$ $[\gcd(r, N) = 1]Andtracegeometricalguidestothatclasses, comparethevisualdistributionoftheprimeswiththeperm$ $[\varepsilon = N - 2\pi m]Weanalicewhatmodulesproducemorepersistentlines.$